NAME: ____________________ ID: __________

CSC 343 H5S – SOLUTIONS

Day Class 01 | Instructor: Michael Liut

DURATION: 2 hours

University of Toronto | Midterm Examination | March 5th, 2018

Read all instructions before starting the exam

question(s)	mark	out of
1 – 10		15
11		6
12		3
13		3
14/15		3
misc.		n/a
total		30

This test paper includes 14 pages and a total of 14 questions; 10 true-or-false questions and 4 questions requiring a written answer. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions :

1. The answers to the written questions (Questions 11-[14/15]) must be answered in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked.
2. Read all questions before starting the exam – some are easier than others.
3. The use of the CASIO FX-991 is permitted, but highly discouraged. No questions herein require a calculator.
4. **This is a closed book exam.** No memory aids, notes, or textbooks of any kind are allowed during the test. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
5. The students are not allowed to be involved in any communication of any kind concerning the questions and answers of this exam with anybody except the invigilator(s) or the instructor. Any attempt of such communication will be considered a case of academic dishonesty.
6. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked.
7. Documents to be returned: this questionnaire, the cheat sheet and all scrap paper if used. All of these documents must bear the student's name and number. Only the face page of the questionnaire needs to bear the student's name and number.

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Question 1 – 10 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example.

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word "True" or the word "False". The negative marking scheme is not used. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. You must justify your answer.

I. General Knowledge (15 marks)

Question 1 [1.5 marks] A relation `R(A,B,C)` may have at most one (minimal) key (not superkeys).

A. True.
$\Longrightarrow$B. False.

FALSE. Each attribute `A`, `B`, and `C` could individually be a minimal key. Thus, the answer would be 3.

Marking Scheme:
[0.5 marks for identifying this statement is FALSE]
[1 mark for justifying that the statement is FALSE]
[mark of 0 if student identifies the answer as being TRUE]

This question maps to: Intended Learning Outcome 1a.

Question 2 [1.5 marks] A weak entity-set is an entity-set that cannot be uniquely identified by its attributes alone; therefore, it must use its primary key in conjunction with a foreign key to uniquely identify tuples.

A. True.
$\Longrightarrow$B. False.

FALSE. Weak entity-sets do not have a primary key based on its own attributes, the foreign key is part of its primary key not in conjuction to it.

Marking Scheme:
[0.5 marks for identifying this statement is FALSE]
[1 mark for justifying that the statement is FALSE]
[mark of 0 if student identifies the answer as being TRUE]

This question maps to: Intended Learning Outcome 1a.

Question 3 [1.5 marks] When speaking of referential integrity, the dependant table is denoted by the creation of a Foreign Key in a table. The table being referenced would, therefore, be the parent.

⟹A. True.
B. False.

TRUE. Students must show an example of this.

Marking Scheme:
[0.5 marks for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 1a and 1b.

Question 4 [1.5 marks] Let relation `R` be a bag over the attributes `A,B`. Now let's say that `A` is our primary key for the relation `R`. Thus, `R` is necessarily a set.

⟹A. True.
B. False.

TRUE. Students must show an example of this.

Marking Scheme:
[0.5 marks for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 1a.

Question 5 [1.5 marks] A weak entity-set `E` has a relationship with a supporting entity-set `R`. The relationship between `E` and `R` must be one-to-one.

A. True.
⟹B. False.

FALSE. Their relationship could be one-to-many.

Marking Scheme:
[0.5 marks for identifying this statement is FALSE]
[1 mark for justifying that the statement is FALSE]
[mark of 0 if student identifies the answer as being TRUE]

This question maps to: Intended Learning Outcome 1a.

Question 6 [1.5 marks] Arity is the number of arguments/operands that a function takes in. Cardinality is a measure of the number of elements of the set. More specifically, cardinality is the number of rows and degree/arity is the number of columns in a relation's instance.

⟹A. True.
B. False.

TRUE. Students may justify by definition or illustrate through example.

Marking Scheme:
[0.5 marks for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 1a.

Question 7 [1.5 marks] Set union is idempotent, i.e. $S \cup S = S$; while under bag law it is not, i.e. $S \cup S \mathrel{!}= S$. Thus, in general, it is safe to assume that intersection is also NOT idempotent; i.e. $S \cap S \mathrel{!}= S$.

A. True.
⟹B. False.

FALSE. Intersection is idempotent for sets and bags. Students may justify by definition or illustrate through example.

Marking Scheme:
[0.5 marks for identifying this statement is FALSE]
[1 mark for justifying why the statement is FALSE]
[mark of 0 if student identifies the answer as being TRUE]

This question maps to: Intended Learning Outcome 2a and 2b.

Question 8 [1.5 marks] Given two relations, `R(A,B)` and `S(B,C)`, $\sigma_{A=B}(R \times S) = (R \bowtie_{A=B} S)$.

$\Longrightarrow$A. True.
B. False.

TRUE. Students may justify by definition or illustrate through example.

Marking Scheme:
[0.5 mark for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 2a and 2b.

Question 9 [1.5 marks] Given two relations, `Emps` and `WorksIn`, if the following `SQL Query` is executed, then its result is undoubtedly the `Output` relation.

Emps:

Emp	Sal	Mgr
Xin	100	Pat
Jen	200	Jen
Pat	100	Meg
Meg	150	Meg
Joe	110	Ami
Ami	200	Bob

WorksIn:

Dept	Emp
Toy	Xin
Car	Bob
Car	Meg
Art	Pat
Art	Meg

$\Longrightarrow$A. True.
B. False.

SQL Query

```
SELECT *
FROM Emps,
    (select Emps.emp, max(sal) as sal
    from Emps, WorksIn
    where Emps.emp = WorksIn.emp
    group by Emps.emp) as foo
WHERE Emps.mgr = foo.emp
```

Output:

Emp	Sal	Mgr	Emp	Sal
Xin	100	Pat	Pat	100
Pat	100	Meg	Meg	150
Meg	150	Meg	Meg	150

TRUE. Students may justify by definition or illustrate through example.

Marking Scheme:
[0.5 mark for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 2a and 2b.

Question 10 [1.5 marks] Given the two relations in Question 9, `Emps` and `WorksIn`, if the following `SQL Query` is executed, then its result is undoubtedly the `Output` relation.

SQL Query

```
SELECT E1.Emp
FROM Emps E1 full outer join Emps E2 on E1.Emp = E2.Mgr
GROUP BY E1.Emp, E2.Emp
HAVING  sum(E1.Sal) >= sum(E2.Sal)
ORDER BY E1.Emp
```

Output:

Emp
Ami
Jen
Meg
Meg
Pat

⟹A. True.
B. False.

TRUE. Students may justify by definition or illustrate through example.

Marking Scheme:
[1 mark for identifying this statement is TRUE]
[1 mark for justifying why the statement is TRUE]
[mark of 0 if student identifies the answer as being FALSE]

This question maps to: Intended Learning Outcome 2a and 2b.

Questions 11 – 13 are questions that require a written answer (some in the form of an SQL query). The answers are to be provided in the space provided below each question.

II. SQL (12 marks)

Question 11 [**6 marks**] Consider the schema below about movies and actors. Keys are underlined.

- `Actor(aid, name)`
- `Movie(mid, title, year, genre, length)`
- `Roles(aid, mid, character)`

Write an SQL query which finds all actors who did not star in any movie (i.e. play a character) in the year 2000. Only return the actor's name and actor ID.

Answer:

```
SELECT aid, name
FROM Actor
WHERE aid NOT IN
        (SELECT roles.aid
        FROM  Actor, Roles, Movie
        WHERE year = 2000 AND actor.aid = roles.aid
        AND roles.mid = movie.mid);
```

Marking Scheme:

[1 marks given to the outer query: 1 mark for correct SELECT attributes and FROM table]

[3 marks given to the subquery: 1 mark for use of correct tables in FROM, 2 marks for use of correct condition in WHERE]

[2 marks for correct subquery join: 1 mark for use of NOT IN, 1 mark for having the correct joining factor in the outer WHERE and subquery's SELECT]

This question maps to: Intended Learning Outcome 2a.

Question 12 [**3 marks**] Consider the following `CREATE TABLE` definition:

```
CREATE TABLE Midterm(
    A INT NOT NULL,
    B INT NOT NULL,
    C INT NOT NULL,
    PRIMARY KEY (A),
    FOREIGN KEY (B) REFERENCES Midterm(A) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (C) REFERENCES Midterm(A) ON DELETE CASCADE ON UPDATE RESTRICT
);
```

Now consider that the table `Midterm` has the following entries:

A	B	C
3	3	3
4	3	3
5	4	3

What is the result of the following SQL statement:

```
DELETE
FROM Midterm
WHERE A = 3
```

Answer:

This will result in an empty table due to the dependencies. Students may state this finding, or, alternatively, show an empty table.

NOTE: Those who completed this midterm under the impression that read:

```
FOREIGN KEY (B) REFERENCES Person(A) ON DELETE CASCADE ON UPDATE CASCADE,
FOREIGN KEY (C) REFERENCES Person(A) ON DELETE CASCADE ON UPDATE RESTRICT
```

whereby it read `Person` instead of `Midterm`. An acceptable alternative solution would be:

A	B	C
4	3	3
5	4	3

Marking Scheme:

[3 marks are awarded for the correct solution with proper justification.]
This question maps to: Intended Learning Outcome 1b and 2a.

Question 13 [**3 marks**] Recall the **Cartesian Product** of two relations R and S, denoted $R \times S$, where R and S are sets: $R \times S = \{(r,s) : (r \in R) \text{ and } (s \in S)\}$. Is the Cartesian product operation commutative? That is, in general, is $(R \times S) = (S \times R)$? Prove/Disprove.

Answer:

$$R \times S = (r,s) : (r \in R) and (s \in S)$$

Note: The Cartesian product of two sets is a set, and the elements of that set are ordered pairs. In each ordered pair, the first component is an element of `R`, and the second component is an element of `S`.

For example, let:
$R = 1, 2, 3$ and $S = (w, x), (y, z)$,
$R \times S = (1, 2, (w, x)), (1, 2, (y, z)), (3, (w, x)), (3, (y, z))$.

If `R` and `S` are both finite sets, then $|R \times S| = |R| \cdot |S|$ as there are $|R|$ choices for the first component of each ordered pair and also $|S|$ choices for the second component of the ordered pair. Therefore, Cartesian product is **not commutative** for the sets `R` and `S`. This is justified in the example above proving that $(R \times S)! = (S \times R)$.

Marking Scheme:
[2 marks for a correct proof (i.e. with example)]
[1 mark for a proper explanation of why or why not]

This question maps to: Intended Learning Outcome 2b.

Questions 14 – 15 are questions that involve the translation of SQL queries into relational algebra. Each of these questions relate to the following schema:

- `Suppliers(sid:integer, sname:string, address:string)`
- `Parts(pid:integer, pname:string, colour:string)`
- `Catalog(sid:integer, pid:integer, cost:real)`

Please note, in the schema above, keys for each relation are denoted by an underlined. Additionally, the `Catalog` relation lists the prices charged for parts by `Suppliers` (i.e. cost).

III. Relational Algebra (3 marks)

Pick ***only one*** to answer: Question 14 or Question 15. ***Do not answer both.***

Question 14 [**3 marks**] Given the following SQL query:

```
SELECT C.sid
FROM   Catalog C
WHERE NOT EXISTS (SELECT P.pid
                  FROM   Parts P
                  WHERE NOT EXISTS (SELECT CG.sid
                                    FROM   Catalog CG
                                    WHERE CG.sid = C.sid  AND
                                          CG.pid = P.pid));
```

Write the relational algebra translation:

Answer:

$$((\pi_{sid,pid}\ Catalog)\ /\ \pi_{pid} Parts))$$

Marking Scheme:

[1 mark for identifying the correct operator]
[1 mark for identifying the correct operands (i.e. relations/tables)]
[1 mark for selecting the appropriate data]

This question maps to: Intended Learning Outcome 2b.

Question 15 [**3 marks**] Given the following SQL query:

SELECT `S.sid`
FROM `Suppliers S`
WHERE `S.address` = '1280 *Main Street*' OR
`S.sid` **IN** (**SELECT** `C.sid`
FROM `Parts P, Catalog C`
WHERE `P.pid` = `C.pid` AND `p.colour` = 'red'

Write the relational algebra translation:

Answer:

$$\rho(R_1,\ \pi_{sid}((\pi_{pid}\ \sigma_{colour='red'}\ Parts)\ \bowtie\ Catalog))$$

$$\rho(R_2,\ \pi_{sid}\ \sigma_{address='1280\ Main\ Street'}\ Suppliers)$$

$$R_1\ \cup\ R_2$$

Marking Scheme:

[1 mark for identifying the correct operators]
[1 mark for identifying the correct operands (i.e. relations/tables)]
[1 mark for selecting the appropriate data]

This question maps to: Intended Learning Outcome 2b.

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